

Ex. No. : 3	Smart Lighting
Date: 21-07-2025	

Aim

To design and implement a Smart Lighting System using an Arduino microcontroller.

Algorithm

Step 1. Start the Arduino program.

Step 2. Define LED pins, Set pins 13, 12, and 11 as output using the pinMode() function.

Step 3. In the loop() function:

a. Turn ON LED connected to pin 13, wait for 2 seconds, then turn it OFF.

b. Turn ON LED connected to pin 12, wait for 2 seconds, then turn it OFF.

c. Turn ON LED connected to pin 11, wait for 2 seconds, then turn it OFF.

Step 4. The loop repeats indefinitely, creating a continuous sequential lighting effect.

Step 5. Observe the sequence to confirm that lights operate automatically and cyclically.

Source Code

SKETCH.IO:

```

void setup() {
  // put your setup code here, to run once:
  pinMode(13, OUTPUT);
  pinMode(12, OUTPUT);
  pinMode(11, OUTPUT);
}

void loop() {
  // put your main code here, to run
  repeatedly:  digitalWrite(13, HIGH);
  delay(2000);  digitalWrite(13, LOW);
  digitalWrite(12, HIGH);  delay(2000);
  digitalWrite(12, LOW);  digitalWrite(11,
HIGH);  delay(2000);
  digitalWrite(11, LOW);
}

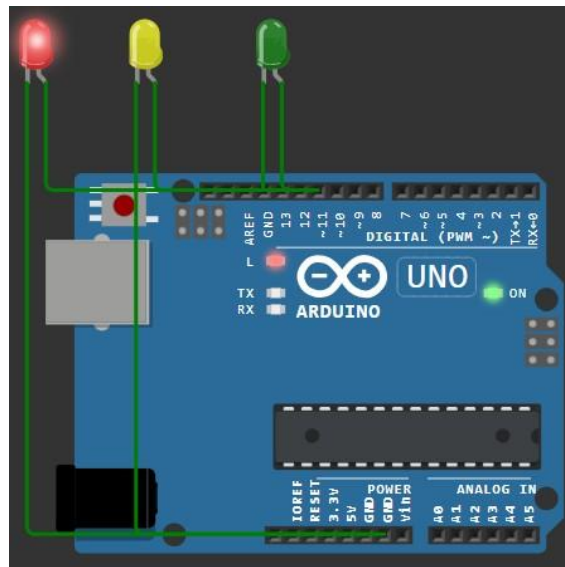
```

Circuit connections

For three LEDs connected externally:

LED Pin	Arduino Pin	Connection
LED1 Anode (A) 13		Connect via resistor (220Ω) to pin 13
LED1 Cathode (C) GND		Connect to GND
LED2 Anode (A) 12		Connect via resistor to pin 12
LED2 Cathode (C) GND		Connect to GND
LED3 Anode (A) 11		Connect via resistor to pin 11
LED3 Cathode (C) GND		Connect to GND

Output



Result

The Arduino successfully controlled three LEDs, demonstrating a basic smart lighting automation system